INCH-POUND

MIL-R-55182/6M
20 December 1989
SUPERSEDING
MIL-R-55182/6L
3 July 1985

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM,

ESTABLISHED RELIABILITY,

STYLE RNR70[1]/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RNR70[1]/, established reliability, film, fixed resistors.

2. APPLICABLE DOCUMENT

* 2.1 Government document.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-55182 - Resistors, Fixed, Film, Established Reliability, General Specification for.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

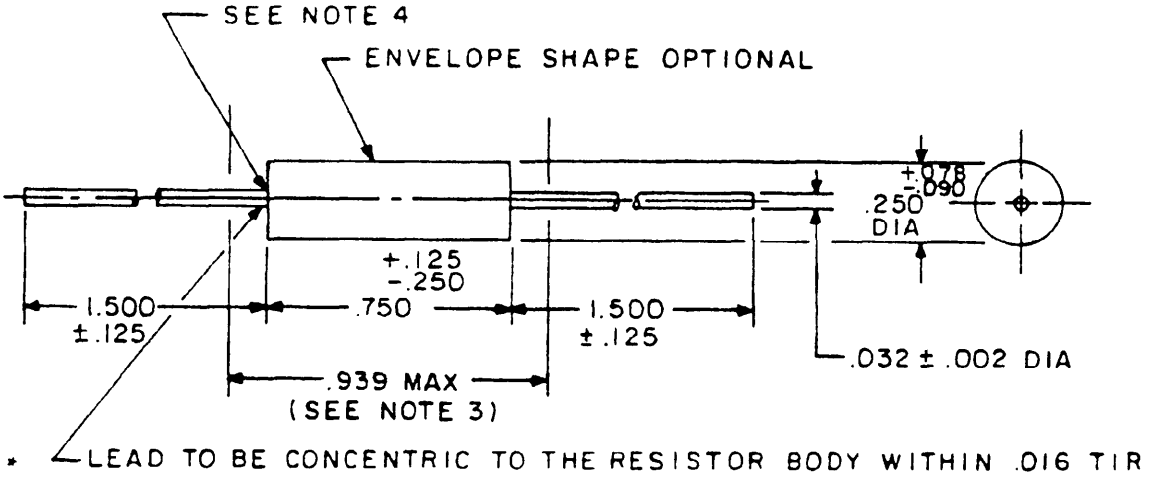
1/ Third letter is variable, dependent upon lead material or capability.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5302 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The maximum length is "clean lead" to "clean lead".
4. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter.
5. Lead length for new design and tape and reel packaging shall be 1.00 +0.625, -0.000 inch (25.4 +15.80, -0.00 mm).
* 6. Lead concentric tolerance is to be measured at the point of lead egress from the resistor body.

Inches	mm
.002	0.05
.016	0.41
.032	0.81
.078	1.98
.090	2.29
.125	3.18
.250	6.35
.750	19.05
.939	23.85
1.500	38.10

FIGURE 1. Style RNR70 2/ resistor.

2/ See footnote 1/.

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3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-55182, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Characteristics. Style RNR70 is available in characteristics C, E, J, H, and K.

3.2.2 Terminal types. Terminal types RNR, RNC, and RNN are available in all characteristics. Terminal type RNR is inactive for design with characteristics H, J, and K.

3.3 Power rating. The power rating shall be 0.50 watt (see 6.3).

3.4 Voltage rating. The continuous working voltage shall not exceed 350 volts.

3.5 Resistance. Minimum and maximum resistance values shall be as follows:

Minimum resistance:

Characteristic:

H, J, and K 3/ - - - - - - - - - - - - - - - - 1.0 ohm

C and E - 20.0 ohms

Maximum resistance:

Characteristic:

C and E - 7.5 megohms

H, J, and K - - - - - - - - - - - - - - - - - - 15.0 megohms

3.6 Voltage coefficient (applicable to resistors of 1,000 ohms and above). The voltage coefficient shall not exceed ±0.005 percent per volt.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-55182.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-55182.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-R-55182 are applicable to this specification.

3/ For resistance tolerances D and F only. For resistance tolerance B, the minimum resistor value shall be 10.0 ohms.

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* 6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Power rating at 70°C. The power rating at 70°C for 2,000 hours of life test has been established at 0.75 watt, 500 volts maximum voltage. It should be noted that the failure rate level is established at the condition of 125°C, .50 watt and 10,000-hour life test duration.

6.4 Supersession data. This specification supersedes MIL-R-10509/12(USAF) in the following manner:

Characteristic in MIL-R-55182 FR level M	Supersedes characteristic in MIL-R-10509/12(USAF)
K	D
C or E	G (up to 1.5 megohms)
H	C
J	E

6.5 MIL-R-10509 substitution data. Resistors of this specification, regardless of their failure rate designation, are substitutes for resistors of the same resistance value, tolerance and appropriate characteristics specified in the inactivated specification MIL-R-10509/3.

TABLE I. Characteristic and terminal substitution.

MIL-R-10509 style RN70	MIL-R-55182/6 substitute characteristic	MIL-R-55182/6 substitute terminal
Inactive characteristic		
C	H	RNC
D	K	RNC
E	J	RNC
G	C	RNC and RNR

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1177)

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